Using Predictive Analytics to Identify Loan Default Risk

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I. EXECUTIVE SUMMARY

This project focuses on predicting whether a borrower will default on a loan using financial and loan-related data. Identifying high-risk borrowers is a key objective for lenders, as undetected defaults can lead to significant financial losses. By developing predictive models, lenders can improve risk assessment and make more informed lending decisions.

The dataset used in this analysis contains over 32,000 loan observations and includes variables such as borrower income, employment length, home ownership status, loan amount, interest rate, loan grade, and loan-to-income ratio. The target variable, loan_status, indicates whether a borrower defaulted (1) or did not default (0). This dataset provides a realistic foundation for both statistical analysis and predictive modeling.

A structured approach was used to analyze the data. Initial exploratory data analysis (EDA) was conducted to understand distributions, identify missing values, and uncover relationships between variables and default behavior. Data preprocessing steps included handling missing values, encoding categorical variables, and preparing the dataset for modeling.

Three models were developed: Logistic Regression, Random Forest, and Gradient Boosting. Logistic Regression was used as a baseline model to examine the relationship between independent variables and default risk. Random Forest and Gradient Boosting were then implemented to improve predictive performance by capturing nonlinear relationships in the data. Model performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC curves, with particular emphasis on recall for the default class.

The results show that tree-based models outperform Logistic Regression. Logistic Regression achieved approximately 86% accuracy but had lower recall, identifying only about 54% of default cases. Random Forest performed best overall, achieving approximately 93% accuracy and a recall of 72%, significantly improving the detection of high-risk borrowers. Gradient Boosting also performed well, with slightly lower recall compared to Random Forest.

Feature importance analysis from the Random Forest model identified loan_percent_income, or loan burden relative to income, as the most influential variable, followed by income, interest rate, and loan amount. These results indicate that loan affordability and borrower financial capacity are the primary drivers of default risk.

Based on these findings, Random Forest is recommended as the final model due to its strong predictive performance and ability to effectively identify default cases. This model can be used by lenders to improve credit risk assessment and reduce exposure to potential losses.

II. INTRODUCTION AND BUSINESS CONTEXT

The objective of this project is to develop a model that can predict whether a borrower will default on a loan based on their financial and loan characteristics. Credit default prediction is a fundamental problem in financial risk management, as lenders must assess the likelihood that a borrower will fail to repay a loan before issuing credit. The goal of this analysis is to build and evaluate models that accurately classify borrowers as either high-risk borrowers likely to default or low-risk borrowers unlikely to default.

This problem is important because inaccurate risk assessment can have significant financial consequences. If lenders fail to identify high-risk borrowers, they may approve loans that ultimately result in default, leading to financial losses. On the other hand, overly conservative lending decisions may result in missed opportunities for profitable loans. Therefore, improving the ability to distinguish between default and non-default borrowers can enhance both risk management and profitability.

This analysis is guided by several key research questions:

- Which borrower and loan characteristics are most strongly associated with default risk?
- How effectively can different models predict whether a borrower will default?
- Do machine learning models outperform traditional statistical approaches in this context?
- Which model provides the best balance between accuracy and the ability to detect default cases?

The dataset used in this project was obtained from Kaggle's Credit Risk Dataset by user Lao Tse. It contains over 32,000 observations of loan-level data and includes a variety of features related to borrower financial status and loan attributes, such as income, employment length, home ownership, loan amount, interest rate, loan grade, and loan-to-income ratio. The target variable, loan_status, indicates whether a borrower defaulted (1) or did not default (0).

This dataset provides a realistic representation of lending scenarios and allows for both statistical analysis and predictive modeling. By applying multiple modeling techniques, this analysis aims to identify key drivers of default risk and determine which model best supports credit risk decision-making.

III. EXPLORATORY DATA ANALYSIS

The dataset contains 32,581 observations and 12 variables, including borrower characteristics, loan information, and credit history indicators. The target variable is loan_status, where 0 represents non-default and 1 represents default. The dataset includes both numerical variables, such as income, employment length, loan amount, interest rate, and loan percent of income, as well as categorical variables, such as home ownership, loan intent, loan grade, and prior default status.

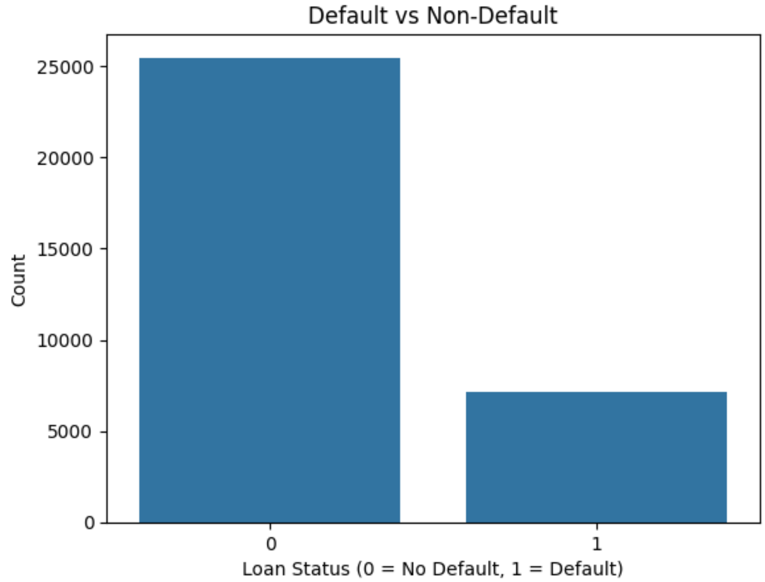
Initial data inspection showed that the dataset was mostly complete. Only two variables contained missing values: person_emp_length had 895 missing values, and loan_int_rate had 3,116 missing values. These missing values were addressed during preprocessing.

Table 1: Data structure and missing values output

Variable	Data Type	Missing Values
person_age	int64	0
person_income	int64	0
person_home_ownership	object	0
person_emp_length	float64	895
loan_intent	object	0
loan_grade	object	0
loan_amnt	int64	0
loan_int_rate	float64	3,116
loan_status	int64	0
loan_percent_income	float64	0
cb_person_default_on_file	object	0
cb_person_cred_hist_length	int64	0

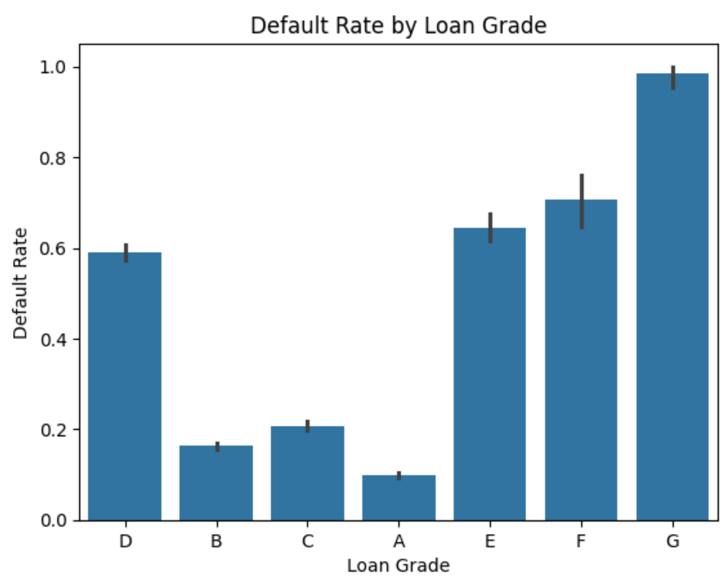
The target variable is imbalanced, with approximately 78.18% non-default observations and 21.82% default observations. This imbalance is important because a model could achieve high accuracy by predicting mostly non-default cases while still missing many actual defaults. As a result, model evaluation later in the report focuses not only on accuracy, but also on precision, recall, F1-score, confusion matrices, and ROC curves.

Figure 1: Default vs non-default



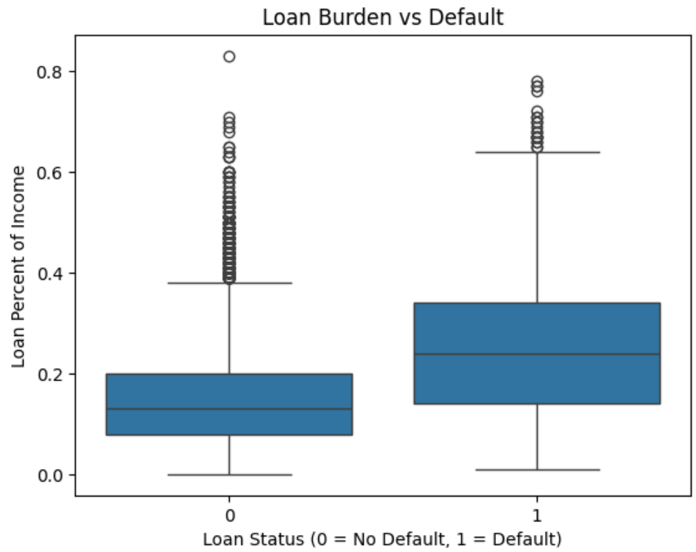
Several important relationships were identified during EDA. First, default rates increase sharply as loan grade worsens. Borrowers with stronger grades, such as A and B, have much lower default rates, while borrowers with weaker grades, especially E, F, and G, show substantially higher default rates. This suggests that loan grade is an important risk indicator.

Figure 2: Default rate by loan grade



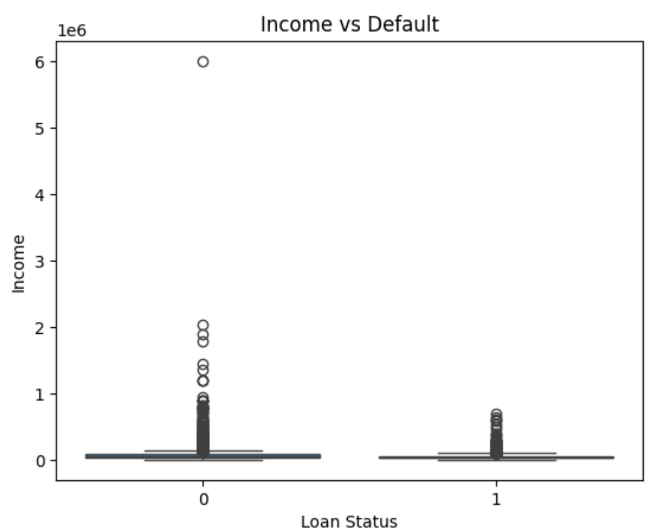
Second, loan burden relative to income appears strongly related to default. Borrowers who default tend to have higher loan_percent_income, meaning the loan represents a larger share of their income. This suggests that affordability is a major driver of repayment risk.

Figure 3: Loan burden vs default



Income also appears relevant, although the relationship is less clear due to extreme outliers. Borrowers who default generally appear to have lower income levels, but income alone is not as strong visually as loan burden or loan grade. This suggests that income is most useful when considered alongside loan amount, interest rate, and loan percent of income.

Figure 4: Income vs default



Overall, the EDA suggests that default risk is most closely connected to loan grade, loan burden, borrower income, and other financial capacity measures. These findings informed the preprocessing and modeling stages by identifying variables likely to be important predictors of loan default.

IV. METHODOLOGY

Before model development, the dataset was prepared to ensure that it could be used effectively in statistical and machine learning models. The preprocessing process included handling missing values, addressing outliers, encoding categorical variables, splitting the data into training and testing sets, and scaling features where appropriate.

Missing values were present in person_emp_length and loan_int_rate. Since both variables are numerical and the dataset contains outliers, missing values were filled using the median rather than the mean. This approach reduces the influence of extreme values while preserving the observations. The person_emp_length variable also contained unrealistic values, so employment lengths greater than 60 years were capped at 60.

Categorical variables were converted into numerical format using dummy variable encoding. These variables included person_home_ownership, loan_intent, loan_grade, and cb_person_default_on_file. After encoding, the dataset expanded from 12 original columns to 23 columns. The target variable was loan_status, and the remaining variables were used as predictors. The data was then split into training and testing sets using an 80/20 split.

The dataset already included loan_percent_income, which measures the loan amount relative to borrower income. This variable was retained because it directly captures borrower loan burden. No additional complex feature engineering was performed because the dataset already contained meaningful borrower, loan, and credit history variables.

Three models were selected: Logistic Regression, Random Forest, and Gradient Boosting. Logistic Regression was included as an interpretable baseline model because the dependent variable is binary and the model provides coefficients, standard errors, z-values, and p-values. Random Forest was selected because it is an ensemble method that combines multiple decision trees and can capture nonlinear relationships. Gradient Boosting was included as a second tree-based ensemble model to compare whether sequential tree-building improved predictive performance.

Because this is a classification problem, model performance was evaluated using accuracy, precision, recall, and F1-score. Since the dataset is imbalanced, recall for the default class was emphasized. Recall measures the percentage of actual defaults correctly identified by the model, which is especially important because missed defaults represent potential financial losses. Confusion matrices and ROC/AUC were also used to compare model performance across classification outcomes and thresholds.

After the initial model comparison, Random Forest was selected for hyperparameter tuning because it produced the strongest balance of accuracy, recall, and F1-score. GridSearchCV was used to test different values for n_estimators, max_depth, and min_samples_split, with recall used as the scoring metric. The best tuned model used 200 trees, no maximum depth restriction, and a minimum split size of 2.

V. RESULTS AND MODEL COMPARISON

Three models were evaluated for loan default prediction: Logistic Regression, Random Forest, and Gradient Boosting. Because the target variable is imbalanced, model performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC/AUC. Recall for the default class was emphasized because missed defaults represent the most financially important error in a lending context.

Table 2: Model performance comparison

Model	Accuracy	Precision Default	Recall Default	F1 Score Default	AUC
Logistic Regression	0.86	0.76	0.54	0.64	0.87
Random Forest	0.93	0.94	0.72	0.82	0.93
Gradient Boosting	0.92	0.92	0.69	0.79	0.93

The results show that Random Forest was the strongest overall model. It achieved the highest accuracy and the best recall and F1-score for the default class. Gradient Boosting also performed well, but slightly underperformed Random Forest in recall and F1-score. Logistic Regression was useful as an interpretable baseline but was weaker at identifying default cases.

Figure 5: Gradient boosting confusion matrix

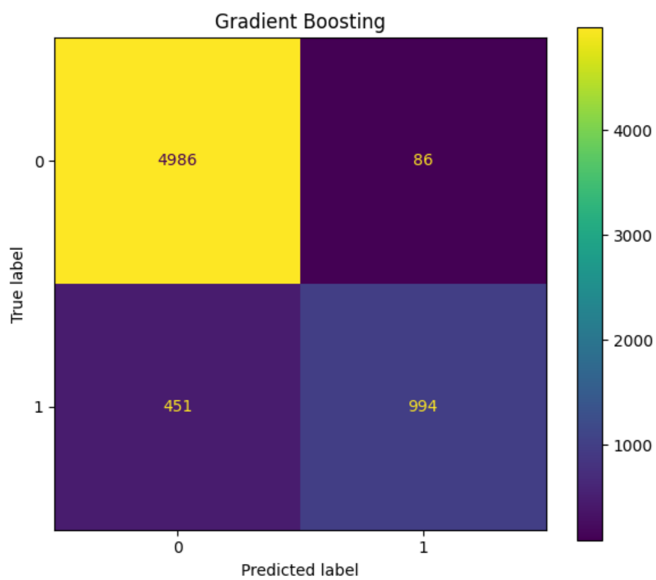


Figure 6: Logistic regression confusion matrix

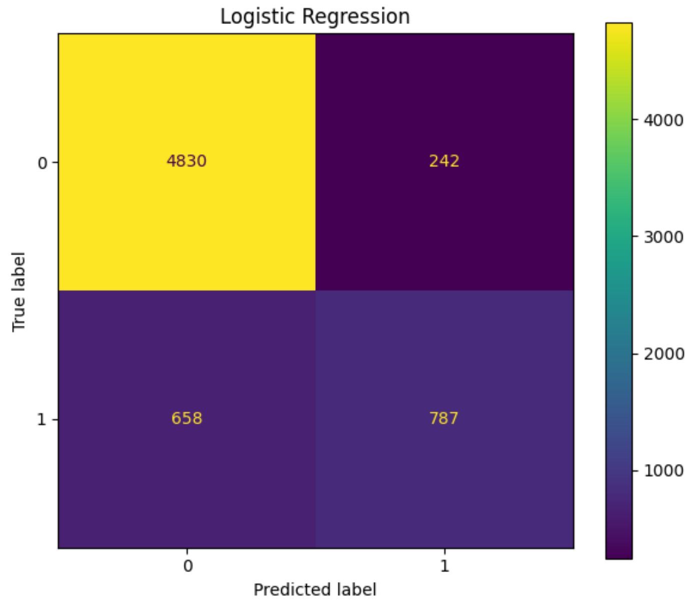
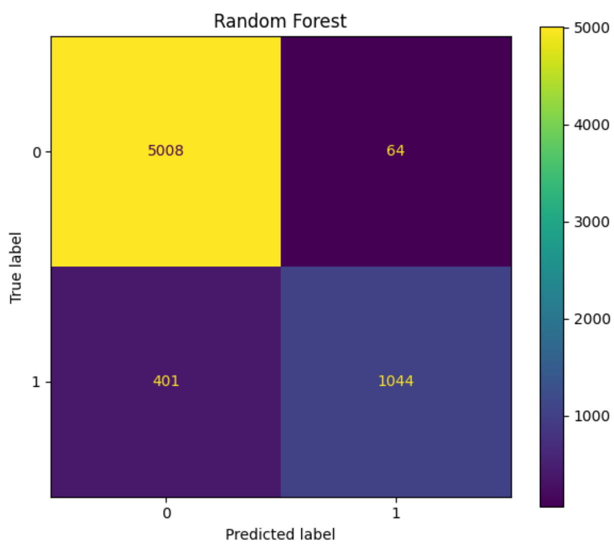
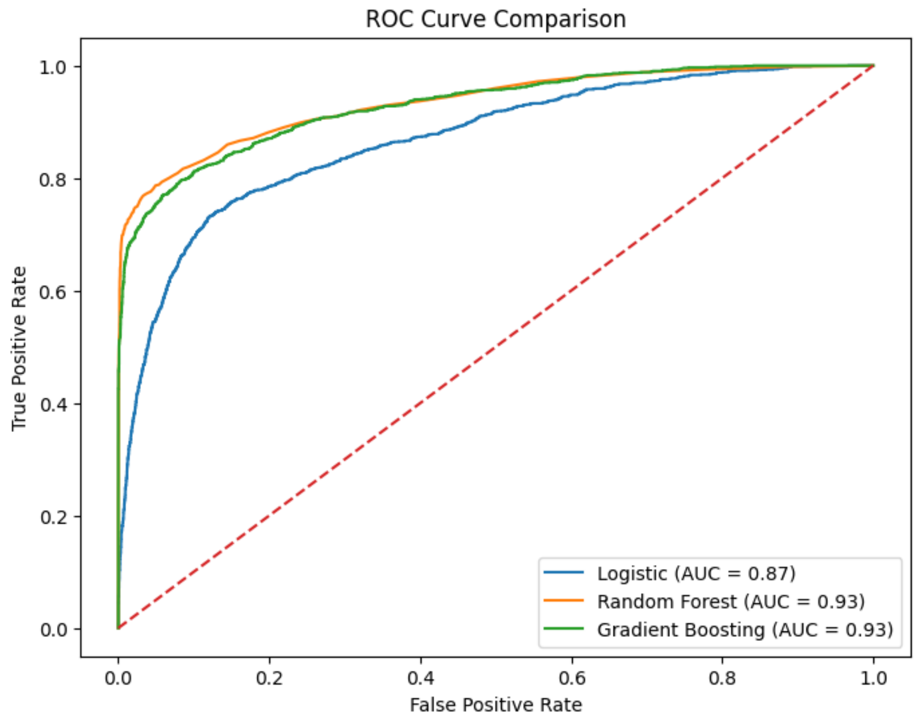


Figure 7: Random forest confusion matrix



The confusion matrices further support these findings. Logistic Regression missed 658 default cases, while Random Forest missed 401 and Gradient Boosting missed 451. Since false negatives represent borrowers who defaulted but were predicted as non-default, reducing this number is especially important. Random Forest produced the lowest number of missed defaults, making it the strongest model from a credit risk perspective.

Figure 8: ROC curve comparison



The ROC curve comparison shows that Random Forest and Gradient Boosting both achieved an AUC of approximately 0.93, compared to 0.87 for Logistic Regression. This indicates that the tree-based models were better at distinguishing between default and non-default borrowers across different classification thresholds.

After the initial comparison, Random Forest was selected for hyperparameter tuning because it had the best balance of accuracy, recall, and F1-score. GridSearchCV tested different values for n_estimators, max_depth, and min_samples_split, using recall as the scoring metric. The best model used 200 trees, no maximum depth restriction, and a minimum split size of 2.

Table 3: Tuned random forest hyperparameters

Hyperparameter	Best Value
n_estimators	200
max_depth	None
min_samples_split	2

Tuning produced only a slight improvement over the baseline Random Forest, suggesting that the original model was already well-specified. Overall, Random Forest was selected as the final model because it provided the strongest combination of predictive accuracy and default detection. While Gradient Boosting had similar AUC performance, Random Forest identified more actual default cases and produced fewer false negatives.

VI. BUSINESS INSIGHTS AND RECOMMENDATIONS

The Random Forest model provides business value by helping lenders identify borrowers with elevated default risk before loans are approved. The model's strongest predictor was loan_percent_income, or loan burden relative to income, indicating that affordability is central to credit risk. Borrowers whose loan amount represents a high share of income should receive additional review because they may have less flexibility to manage repayment obligations.

Table 3: Random forest feature importance

Rank	Feature	Importance	Business Meaning
1	loan_percent_income	0.222	Loan burden relative to income
2	person_income	0.150	Borrower financial capacity
3	loan_int_rate	0.122	Risk-based cost of borrowing
4	loan_amnt	0.080	Size of loan obligation
5	person_home_ownership_RENT	0.078	Housing/financial stability
6	person_emp_length	0.059	Employment stability
7	loan_grade_D	0.053	Higher-risk loan grade
8	person_age	0.049	Borrower demographic factor
9	cb_person_cred_hist_length	0.038	Credit history length
10	person_home_ownership_OWN	0.018	Housing/asset stability

The feature importance results reinforce the main business insight from the model: affordability and borrower financial capacity are the strongest indicators of default risk. Loan burden, income, interest rate, and loan amount should therefore receive close attention during underwriting and risk review.

From an implementation standpoint, the Random Forest model should be used as a decision-support tool in the loan approval process. For each applicant, the model can generate a predicted default classification and probability of default. Borrowers with elevated predicted risk could be flagged for manual review, additional documentation, adjusted loan terms, or potential denial. This would allow the lender to make more data-driven decisions while still preserving human judgment in final approval.

The model could also support risk-based pricing and loan structuring. Borrowers with higher predicted default risk may require smaller approved loan amounts, stricter repayment terms, or additional review before an interest rate is finalized. However, this should be applied carefully, since charging higher rates to already risky borrowers could increase repayment stress and potentially raise default risk further.

In addition, lenders should monitor borrowers with high-risk characteristics after loan origination. Applicants with high loan burden, lower income, higher interest rates, or weaker loan grades may benefit from earlier intervention if repayment issues begin to appear. This could include payment reminders, refinancing options, or hardship support before the borrower reaches default.

The model should also be updated over time. Because it was trained on historical data, its usefulness depends on whether future borrower behavior remains similar to past patterns. As economic conditions, interest rates, and borrower behavior change, the model should be retrained and re-evaluated regularly to maintain performance.

Overall, the expected business impact includes better risk assessment, fewer missed defaults, more informed lending decisions, and improved portfolio quality. By combining predictive modeling with human judgment, lenders can make decisions that are both data-driven and practical.

VII. ETHICS AND RESPONSIBLE AI

Because this project focuses on loan default prediction, ethical considerations are especially important. Predictive models used in lending can influence whether borrowers receive access to credit, so the model must be evaluated not only for accuracy, but also for fairness, transparency, and responsible use.

One potential source of bias is that several variables may act as indirect proxies for socioeconomic status. Although the dataset does not explicitly include sensitive demographic variables such as race, gender, or religion, features such as income, employment length, home ownership status, and credit history may still reflect broader economic inequalities. For example, renters or lower-income borrowers may be classified as higher risk, even though these characteristics do not automatically mean a borrower is unwilling or unable to repay.

Another concern is that the model was trained on historical data. If historical lending patterns reflected unequal access to credit or biased approval decisions, the model could learn and reproduce those patterns. This is a common risk in financial machine learning: a model may appear objective because it is data-driven, while still reflecting bias embedded in the training data.

Fairness should therefore be evaluated before any real-world deployment. In addition to overall accuracy and recall, lenders should examine whether the model performs differently across borrower segments such as income groups, home ownership categories, loan purposes, and employment lengths. Attention should be paid to false positives, where borrowers are incorrectly labeled as likely to default. A high false positive rate for a specific group could unfairly limit access to credit.

Privacy and security are also important because the dataset includes sensitive financial information, including income, employment history, loan amount, and credit history. In a real lending environment, this data would need to be protected through access controls, encryption, audit logs, and strict data governance policies. Borrower data should only be used for legitimate lending and risk management purposes.

The model should also be deployed as a decision-support tool rather than a fully automated approval system. A high-risk prediction should trigger further review, not automatic denial. Human oversight is important because the model may not capture all relevant borrower context, such as recent employment changes, savings, co-signers, or temporary hardship.

Responsible deployment would also require ongoing monitoring. Economic conditions, interest rates, and borrower behavior can change over time, which may reduce model accuracy. The model should be retrained and validated regularly using updated data. Lenders should also document model decisions and provide explanations where appropriate, especially when model outputs influence credit access.

Overall, predictive modeling can improve credit risk assessment, but it must be used carefully. A responsible approach should combine model predictions with fairness testing, privacy protection, human oversight, and regular performance monitoring. This ensures that the model supports better lending decisions without creating unnecessary or unfair barriers to credit.

VIII. CONCLUSION AND FUTURE WORK

This project developed and evaluated predictive models to estimate whether a borrower is likely to default on a loan. The analysis began with exploratory data analysis to understand the dataset, identify missing values and outliers, and examine relationships between borrower characteristics and default risk. The strongest patterns showed that default risk is closely related to loan burden, loan grade, interest rate, income, and other measures of borrower financial capacity.

After preprocessing the data, three models were developed: Logistic Regression, Random Forest, and Gradient Boosting. Logistic Regression was useful for interpreting relationships between independent variables and default risk, while the tree-based models were stronger for prediction. Random Forest produced the best overall performance, achieving approximately **93% accuracy** and the strongest recall for the default class. Since identifying actual defaults is the main objective in credit risk modeling, Random Forest was selected as the final model.

The results suggest that affordability is one of the most important drivers of default risk. Feature importance analysis showed that loan_percent_income, income, interest rate, and loan amount were among the most important predictors. From a business perspective, these findings suggest that lenders should pay close attention to whether borrowers can realistically manage repayment obligations relative to their income.

There are several limitations to this approach. First, the model was trained on historical data, so it assumes that future borrower behavior will follow similar patterns. Changes in interest rates, employment conditions, inflation, or broader economic conditions could affect default behavior over time. Second, the dataset does not include all variables that lenders may use in real underwriting, such as credit score, existing debt, savings, payment history, or macroeconomic indicators. Third, although Random Forest performed well, it still missed some default cases, meaning it should support human decision-making rather than replace it.

Future work could improve the model by adding more borrower and economic variables, testing additional algorithms such as XGBoost or LightGBM, and tuning classification thresholds to improve default detection. Fairness testing should also be conducted to ensure that the model does not disproportionately affect specific borrower groups. In addition, the model should be retrained and monitored regularly as new borrower data becomes available.

Overall, this project showed the value of combining statistical modeling and machine learning to support credit risk decisions. It also highlighted the importance of choosing evaluation metrics that match the business problem, since accuracy alone is not enough when the goal is to identify high-risk borrowers.

IX. REFERENCES AND ACKNOWLEDGMENTS

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Tools and Libraries Used

This project was completed using Python in Google Colab. The primary libraries used were pandas, NumPy, matplotlib, seaborn, scikit-learn, and statsmodels.

Pandas and NumPy were used for data loading, cleaning, manipulation, and numerical operations. matplotlib and seaborn were used to create exploratory data visualizations, confusion matrices, ROC curves, and feature importance charts. scikit-learn was used to build and evaluate the Logistic Regression, Random Forest, and Gradient Boosting models, as well as to perform train-test splitting, feature scaling, model evaluation, and hyperparameter tuning. statsmodels was used to estimate the interpretable Logistic Regression model with coefficient estimates, standard errors, z-values, and p-values.

Acknowledgments

The dataset used in this project was sourced from Kaggle's Credit Risk Dataset by Lao Tse. Course materials, lectures, and lab exercises from FIN 835 Predictive Analytics and Machine Learning were used to guide the modeling process, particularly the implementation and evaluation of classification models.

AI assistance was used for limited support with code organization, formatting, citations, and editing report language for clarity. The analysis, model outputs, interpretation of results, and final conclusions were completed and reviewed independently.